Group Proposal Guidelines

For the proposal assignment, you will prepare a short proposal for an innovation related to the theme of ageing. The proposal should be no more than 800 words (excluding references, figures, and any other creative elements you wish to include). This proposal may serve as a foundation for your final poster and presentation. Groups may either build on this proposal or pursue a new, unrelated topic for their poster in the second half of the semester.

Here are some prompts to guide your thinking:

- **Key challenge or gap:** What is a major problem or unmet need related to ageing that you aim to address? You may consider issues faced by seniors or by other stakeholders such as caregivers.
    - How prevalent and significant is this problem?
- **Proposed innovation:** What key technology or innovation are you proposing?
- **Demonstration of interdisciplinary analysis:**
    - **Scientific and technical rationale (STEM):** What are some underlying principles, scientific rationale, or technical features behind your proposed innovation?
    - **Implementation considerations (Biz/Gov):** What are some aspects you need to consider when implementing your solution? For example, feasibility? Sustainability over long-term use?
    - **User perspective (Social)**: How have you considered social factors or user experiences? This could appear in your analysis of the issue, review of existing solutions, or in the design of your innovation.
- **Application of course material:** Have you applied concepts or discussions from lectures and tutorials to support your key ideas where possible?
- **Limitations and challenges**: Can you identify some potential challenges/limitations related to your proposed solution? For example, social impact of your innovation? (if relevant)

Submission:

- Refer to course schedule in syllabus for submission due date.
- They should be submitted to NTULearn via your **cluster** site.
- State your group members' names clearly on the front page of your assignment (as per matriculation card).
- **Three** submissions should be made: one for Turnitin submission, one for group submission, and one for submission of the Declaration of Academic Integrity.
- Only one student needs to submit on behalf of the team. The group submission is viewable by the rest of the group members. The Turnitin submission can only be viewed by the submitter.

Additional notes:

1) When you cite someone's work, you must give due credit and acknowledgment to the original idea or person.
2) **If you use any generative AI (GAI) tools (including Grammarly), you must include a digital paper trail (i.e., your original work with no AI content).** You can include this as an appendix, or provide a link generated by GAI (see for example this 2 min video). See here for how to cite GAI

Frequently Asked Questions

Q: Does our proposed idea need to be situated/applied in Singapore?
A: No, it can be something for the general public outside of Singapore. But more often than not it is easier to design and justify a target solution for a specific target audience. So, you may want to scope your target audience further.

Q: Does our proposed idea need to be based on topics covered directly in class?
A: Not necessarily, but there should be close connections with the lecture content. For example, physical aspects of ageing (osteoporosis, hearing loss, mobility) would be acceptable; diabetes in the general population would not be acceptable.

Q: Do we have to link our idea to class lectures?
A: Yes, this is ideal. The lectures have included many charts, statistics, concepts, and terms that will help you to define your problem and explain the relevance of your solution. The more specific you can be, the better.

Q: Does our proposed idea need to be something tangible?
A: Not necessarily, but it should be something innovative that can solve the identified problem.

Q: Do we need to include citations?
A: If you have consulted sources (including websites) in drafting your proposal, you should include citations in a reference list which is excluded from the word limit. You may want to use citations and references to support claims and statistics that you include in your proposal. It is to make your proposal more convincing by citing credible references.

Q: Does our proposed technological innovation need to be something commercially available?
A: No, it does not. However, it needs to be doable and practical in terms of existing/developing technological readiness and likely cost.

Q: Does our proposed technological innovation need to be something we came up with by ourselves or are we allowed to adapt existing solutions that we identified?
A: Innovation requires new ideas. They do not need to be totally new ideas but should not be the same as existing products or solutions. You may want to think about how you can add value to existing products, have them perform additional tasks, or otherwise make improvements to the design that can better address your identified problem.

Examples

Some examples based on the ageing theme (these are meant for your reference only and we hope this will not limit your ideas):

- **Innovation:** Development of app to help cope with dementia. *(The key challenge is that individuals/families/caregivers may not be able to plan adequate health and care strategies well when a family member is diagnosed with dementia. This tool would give more personalized indicators of needs, as well as activities to slow down decline and calm patients as an anticipatory approach).*

- **Application:** Tech-fair for seniors in senior centres, focusing on diet, health, and cognitive reserve-building. *(The key challenge is that seniors may only access abstract information about maintaining their health, but may not have clear strategies about how to adjust their lifestyles to be healthier. This fair would feature already-existing senior-friendly gadgets for measuring nutritional values of food, improving physical mobility, and exercising cognitive functions with the identified target audience, clear messages, and most effective communication line as important considerations).*

Intended Learning Outcomes (ILO) for this assignment:

ILO 1: (Content) Describe the basic scientific/technical principles of the featured cutting-edge fields/innovations; the kinds of human problems/challenges that these are meant to address; and the additional problems/challenges posed in and by their application.
ILO 2: (Interdisciplinary) Describe and examine connections between three general classes of perspective on the challenges of innovation and application: scientific/technical, business, and humanistic/social scientific perspectives
ILO 3: (Critical thinking) Identify, critique, and evaluate ethical, financial, organizational, legal, social, and cultural components surrounding the application of technological innovations.
ILO 4: (Problem solving) Identify and present possible innovations and solutions for their application that would achieve a consensual balance between scientific-business-social concerns.
ILO 5: (Team) Collaborate effectively on project development in cross-disciplinary teams.